Harini Manivannan

Eragudi, Thuraiyur, Trichy, Tamil Nadu - 621001

+91 94439 49478 | mharini208@gmail.com | linkedin.com/in/harini-manivannan

Professional Summary

Results-driven Electronics and Communication Engineering graduate with hands-on experience in Embedded Systems, IoT, and Microcontroller programming. Skilled in designing real-time monitoring systems using Embedded C, C, and Proteus. Proven ability to integrate wireless communication protocols (GSM, UART) for safety alerts. Passionate about developing efficient, low-power embedded solutions for the automotive and healthcare sectors.

Education

Kongunadu College of Engineering and Technology 2023 – 2027
B.E in Electronics and Communication Engineering; ***CGPA: 7.54/10.0*** *Trichy, Tamil Nadu*

Govt. Girls Higher Secondary School 2022 – 2023
HSC (12th); ***Percentage: 79%*** *Mettupalayam, Tamil Nadu*

AGM Higher Secondary School 2020 – 2021
SSLC (10th); ***Percentage: 100%*** *Eragudi, Tamil Nadu*

Technical Skills

Languages: C, Embedded C, Java (Basics)
Microcontrollers: PIC16F877A, 8051, Arduino, ESP32
Tools & IDEs: MPLAB X IDE, XC8 Compiler, Proteus
Protocols & Communication: UART, I2C, SPI, GSM

Projects

Real-Time E-Vehicle Battery Health Monitoring System | *8051 Microcontroller, ESP32, GSM Module, Embedded C*

- Built an embedded system to monitor critical battery parameters including voltage, current, and temperature in real-time.
- Integrated a GSM module to send automated SMS alerts to mobile devices upon detecting abnormal threshold levels (e.g., overheating, overcurrent).
- Engineered low-power background monitoring logic that functions continuously, even when the vehicle is switched OFF, to enhance battery safety and longevity.

Patient Database Management System | *MySQL, C Programming*

- Developed a centralized database management system to store and retrieve patient records, medical history, and appointments efficiently.
- Implemented secure data storage logic to ensure data integrity and quick retrieval for healthcare providers.
- Reduced manual record-keeping errors and contributed to the automation of appointment management.

Internship

Retech Solutions Jun 2025
IoT Intern *Tambaram, Tamil Nadu*

- Programmed microcontroller-based IoT devices using Arduino and ESP32 platforms for real-time environmental monitoring applications.
- Interfaced various analog and digital sensors with microcontrollers to capture, process, and transmit system data.
- Implemented wireless communication protocols to enable remote data tracking and real-time alerts.

Achievements

- Completed **IoT internship** at Retech Solutions with hands-on experience in sensor interfacing and real-time monitoring.
- Successfully delivered **2 academic projects** in Embedded Systems and Database Management.
- Completed a **Value-Added Course on PCB Design**, gaining proficiency in designing schematics and layouts for embedded applications.